

Evaluating Commercial Facial Analysis Algorithms Pre-Registered Analysis Plan

April 29, 2026

Service Outputs: We record age estimation and liveness prediction outputs as generated by each of three age verification services: Facetec, Privately, and Yoti.¹ We describe these outputs in greater detail below.

In some circumstances, the services provide neither age estimation nor liveness outputs, which we record as “null” outputs, as also detailed below.

Participants may opt out of any treatment condition during testing. We will treat these decisions as a gap in data collection (i.e., no response) and distinct from null responses.

Age estimation outputs:

Table 1: Age estimation output (e) from each service.

	Facetec	Privately	Yoti
Age estimation output (e)	Binary age estimate. Binary age prediction is with respect to four or more age thresholds, ² selected from 13, 16, 18, 21, 25, and 30. ³ Facetec may output a message instead of an age estimate, such as “Let’s try that again.” The	Binary age estimate. Binary age estimation is with respect to three age thresholds, 21, 25, and 30. We have not observed any instances in study data collection of Privately outputting a message instead of	Integer age estimate. We convert integer age estimates to binary age estimates, as described below. Yoti may also output a message instead of an age estimate,

¹ We access these services through first-party smartphone apps and rely on outputs visible in the app user interfaces. Facetec’s Demo app is supported by iOS and Android; Privately’s AgeAI for Retail app is supported by Android only; and Yoti’s API (accessed via Chrome on Android and Safari on iOS) is supported by iOS and Android. We use SDK version 10.0.47 of Facetec’s iOS app and SDK version 10.0.40 of Facetec’s Android app. We use API version 0.15.0 of Yoti’s age antispoofing app. We use version 1.19.4 of Privately’s Android app.

² Because of time constraints, we may not be able to evaluate all six age thresholds for all participants. We will begin with t_a , the highest threshold below the participant’s age (or the highest threshold if none are below the participant’s age), then test at least three additional thresholds, working upward from t_a and then working downward from t_a if the thresholds above t_a are exhausted.

³ Our present understanding, based on developer documentation, is that the Facetec and Privately thresholds are for the age being *greater than or equal to* the threshold. We will confirm this aspect of the systems and, if either is for the age being *greater than but (not equal to)* the threshold in whole years, we will normalize the threshold for comparison across providers.

	service may also fail to output a value, instead continuing to provide instructions such as moving closer or further away. We treat these outputs as null responses.	an age estimate or not outputting a value. If we encounter these behaviors, we will document them and treat them as null responses.	such as “Face not found” or “Multiple faces detected.” We record these as null responses. We have not observed any instances in study data collection of Yoti not outputting a value. If we encounter this behavior, we will document it and treat it as a null response.
--	--	---	---

We will encode binary age estimation outputs as 1 (predicting the user is above the age threshold) or 0 (predicting the user is below the age threshold).⁴

The API that Yoti provided access to for purposes of this research returns an estimated four-year (inclusive) age range (e.g., 21-25). We convert this age range to a binary age estimate by taking the midpoint and applying a buffer, as described below under RQ1.

Liveness outputs:

Table 2: Liveness detection output (*l*) from each service.

	Facetec	Privately	Yoti
Liveness detection output (<i>l</i>)	Binary liveness detection.	N/A (none provided).	Binary liveness detection.

As with age estimation outputs, we will encode binary liveness detection outputs as 1 (detecting that the user is a live human being) or 0 (detecting that the user is not a live human being). There is a degree of ambiguity in the semantics of “liveness,” which we discuss below.

Null outputs:

Table 3: Null output (*n*) from each service.

	Facetec	Privately	Yoti
Null output (<i>n</i>)	Binary null state, mutually exclusive with	N/A (null outputs not observed).	Binary null state, mutually exclusive

⁴ We use this approach for clarity and consistency and to facilitate regression modeling in secondary analysis.

	age estimation and liveness outputs. Facetec may output a message instead of an age estimate and liveness result, such as “Let’s try that again.” These messages sometimes include an explanation, such as image blurriness or poor lighting. We record these as null responses. The service may also fail to output values, instead continuing to provide instructions such as moving closer or further away. We have observed these outputs in data collection and we also treat them as null responses.	We have not observed any instances in study data collection of Privately outputting a message instead of an age estimate or liveness result. Similarly, we have not observed instances of Privately failing to output these values. If we encounter these types of behaviors, we will document them and treat them as null responses.	with age estimation and liveness outputs. Yoti may return a message instead of an age estimate and liveness result, such as “Face not found” or “Multiple faces detected.” We record these as null responses. We have not observed any instances in study data collection of Yoti not outputting a value. If we encounter this behavior, we will document it and treat it as a null response.
--	---	--	--

Null outputs appear to be mutually exclusive with age estimation and liveness outputs across services. If we observe circumstances where apps report these types of issues but nevertheless provide age estimation or liveness outputs, we will report them.

We will encode the binary null state for an instance of data collection as 1 (there was a null output and neither age estimation or liveness output) or 0 (there was age estimation and liveness output). Where we collected data for a particular participant, condition, app, and OS that was null in one instance of data collection and was not null in another (e.g., upon retry), we will discard the null output and use the available age estimation and liveness outputs. We will report counts of instances where we retried data collection and outcomes.

In rare instances, participants have opted out of particular data collection conditions. We will report counts for these opt-out decisions and otherwise omit them from analysis.

RQ1: What is the age predictive performance of age estimation services that rely on biometric facial analysis, when people interact with the services as directed?

We will analyze age estimation outcomes in two different ways.

- 1) **Service-provided age thresholds.** This evaluation characterizes off-the-shelf predictive performance. We will set the classification threshold for system evaluation, t_e , as an age threshold offered by the service provider, t_s .⁵ We will treat each service-provided age threshold as a distinct classification task, and we will report a confusion matrix, the false positive rate, and the false negative rate for each service that provides the threshold. For example, for all three services, we will report these predictive performance measures using the service-provided threshold for whether the participant is at least 21 years old ($t_e = t_s = 21$). We will also report prediction correspondence across the services, as described below.
- 2) **Performance curves.** This evaluation characterizes possible tradeoffs in predictive performance, rather than the configurations that the services recommend. We will fix the classification threshold for evaluation, t_e , separate from the thresholds provided by services. We will use $t_e \in \{13, 16, 18, 21\}$, as these are common legally relevant age thresholds. We will then vary the classification threshold offered by each service provider, t_s , allowing us to characterize tradeoffs in predictive performance. We will report P-R and ROC curves for each service, and we will compute ROC AUC values.

We will use the following definitions in this analysis, in addition to the binary age estimation output e from above. We will discard null responses (i.e., where $n = 1$) for purposes of this analysis.

Table 4: Definitions of analysis values for age estimation predictive performance.

Value	Definition
Age (a)	The participant's ground truth age in years, as self-reported in our survey.
Age point estimate (p)	The midpoint of the four-year (inclusive) age interval returned by Yoti's age estimation service.
Buffer (b)	A buffer that we use to convert point estimates of age (from Yoti) into binary predictions, as described below.
True positive (TP)	The participant's age a is greater than or equal to the evaluation threshold t_e , and a service outputs a binary age estimate e with value 1 when using the service threshold t_s (i.e., $a \geq t_e$ and $e = 1$ with the service configured to t_s).
True negative (TN)	The participant's age a is less than the evaluation threshold t_e , and a service outputs a binary age estimate e with value 0 when using the service threshold t_s (i.e., $a < t_e$ and $e = 0$ with the service configured to t_s).

⁵ Because Yoti does not have a fixed set of classification thresholds, we will use the classification thresholds from Facetec and Privately.

	configured to t_s).
False positive (FP)	The participant's age a is less than the evaluation threshold t_e , and a service outputs a binary age estimate e with value 1 when using the service threshold t_s (i.e., $a < t_e$ and $e = 1$ with the service configured to t_s).
False negative (FN)	The participant's age a is greater than or equal to the evaluation threshold t_e , and a service outputs a binary prediction with value 0 when using the service threshold t_s (i.e., $a \geq t_e$ and $e = 0$ with the service configured to t_s).
Service threshold (t_s)	An age classification threshold supported by a service, which we configure the service to use in an evaluation.
Evaluation threshold (t_e)	An age classification threshold that we fix in an evaluation, for purposes of determining positive and negative classes. The evaluation threshold may be a service threshold, or it may differ when constructing performance curves.
Highest service threshold below age (t_a)	The highest service threshold t_s below the participant's age, or the highest threshold (if none). We use this threshold when determining which thresholds to evaluate for Facetec, as described above.
Mean absolute error (MAE)	For Yoti only, because it provides an age point estimate, we will report MAE defined as: $MAE = \left(\sum_{participants} p - a \right) / (\text{count of participants})$

We will convert Yoti's age point estimates p into binary predictions e by applying a fixed offset buffer $b \in \{2, 3, 5, 7\}$ and computing $e = [p \geq t_s + b]$. For example, if the age point estimate is 27 and $t_s = 21$, we will interpret the output as $e = 1$ (predicted above age threshold) for $b \in \{2, 3, 5\}$ and as $e = 0$ (predicted below age threshold) for $b = 7$. We implement this buffer approach consistent with the service's recommendations,⁶ and we select buffer values that have been recommended by the service or by governments.⁷ We will separately report predictive performance for each buffer value.

To understand the extent of prediction correspondence across services, for each service threshold, we will compute a Fleiss's kappa interrater reliability score, treating each service as a

⁶ <https://www.yoti.com/blog/thoughts-from-our-ceo-july-2025/>

⁷ NIST (USA): https://pages.nist.gov/frvt/reports/aev/fate_aev_report.pdf

KJM (Germany): <https://www.yoti.com/blog/age-estimation-approved-kjm-highest-age-assurance-level-18-adult-content/>

“rater.” We will also create upset plots to visualize the (dis)agreement in predictions across services.

RQ2: How robust is age estimation to a person’s noncompliance with the image capture directions that services provide or adversarial measures that the person may take?

We specify the set of conditions that we will evaluate in the research protocol approved by Princeton University’s IRB, which is attached to this analysis plan.

We will conduct the service-provided age threshold analysis as in **RQ1**, separately reporting results for each condition.

RQ3: How do demographic and personal attributes relate to age predictive performance?

We will associate the following demographic and personal attributes with each participant.

Table 5: Demographic and personal attributes, sources, and definitions.

Attribute	Source	Definition
Age	Survey	The participant’s age in years, stratified as described below.
Race	Survey	The participant’s race, which may involve multiple survey response selections.
Gender identity	Survey	The participant’s gender identity, which may involve multiple survey response selections.
Skin tone	Photographs	The participant’s skin tone on the Monk Skin Tone Scale, which we identify by applying computer vision to baseline condition photographs of the participant.

We will report age predictive performance as in the service-provided age threshold analysis for RQ1, stratifying participants by each attribute and each study condition. We will exclude strata with fewer than three participants, since the data is so sparse as to undermine plausible generalizability.

We will stratify participant ages into the age buckets used in NIST's Face Analysis Technology Evaluation (FATE) Age Estimation & Verification analysis: 13-16 and 17-22 (modeled on a legal threshold of 13 years old), and 14-17 and 18-30 (modeled on a legal threshold of 18 years old). We will also include age buckets of 13-15 and 16-20 (modeled on a legal threshold of 16 years old).

Where participants have selected more than one race or more than one gender identity, we will report results both including the participant's data in all relevant races and gender identities and with the data separate from individual races and gender identities, subject to the strata size reporting limitation above.

In addition to reporting results by individual strata, we will report by joint age and skin tone strata and by joint gender and skin tone strata. This approach is motivated by prior computer vision research demonstrating that these joint attributes may be associated with substantially different classifier performance.

RQ4: What is the liveness predictive performance of these services, when people interact with the services as directed?

For this analysis, we rely on liveness output l (Table 2) and null output n (Table 3). We fix the ground truth liveness f as 1, because all images depict a real person's face, posed and lit consistent with vendor directions, and captured with a high-quality camera.

We will conduct analysis analogous to the service-provided threshold analysis in **RQ1**, reporting confusion matrices, false negative rates (but not false positive rates since there is no negative ground truth data), and correspondence across services.

In addition to conducting this analysis by comparing liveness output to ground truth liveness (i.e., $l \stackrel{?}{=} f$), we will also conduct a version of the analysis that compares combined liveness and not-null output to ground truth liveness (i.e., $l \wedge \neg n \stackrel{?}{=} f$). The former approach directly evaluates the liveness prediction component of systems, and the latter evaluates the usability of age assurance systems (i.e., whether they will allow for age prediction or reject a face).

RQ5: How robust is liveness prediction to noncompliance with service directions or adversarial measures?

There is a degree of ambiguity in mapping the conditions in our research to liveness detection output.

Facetec and Yoti describe their liveness detection as follows.

- Yoti: Liveness detection is intended to determine whether an age or identity scan “is being performed by a real person, and not someone wearing a mask or an automated system.”⁸
- Facetec: Liveness detection is intended to indicate “if the images do not contain a live human.” It is distinct from 3D facemap construction functionality, because “masks and dolls are 3D but are not alive.”⁹

A relevant technical standard, ISO/IEC 30107-1:2023 (Biometric presentation attack detection), defines “liveness detection” as “measurement and analysis of anatomical characteristics or involuntary or voluntary reactions in order to determine whether a biometric sample is being captured from a living subject present at the point of capture.” It defines the related concept of “presentation attack detection” as “automated discrimination between bona-fide presentations and biometric presentation attacks,” and it defines a “biometric presentation attack” as “presentation to the biometric capture subsystem with the goal of interfering with the operation of the biometric system.”

The baseline condition in our research plainly has ground truth liveness $f = 1$ according to these definitions. The full coverage face mask condition also plainly has ground truth liveness $f = 0$. The other conditions are less clear. People with partial facial obstructions or manipulated facial expressions should, at least arguably, be detected as live persons—but they also are people not complying with directions and may be interpreted as people seeking to evade the system. It would be a reasonable design choice for vendors to reject these images as not live, and some may have developed their systems with that goal. Accordingly, we will report sets of results treating these two categories of conditions as live and not live. The following table maps study conditions to the ground truth liveness results that we will use in analysis.

Table 6: Ground truth liveness values by study condition.

Study Condition	Ground Truth Liveness
Baseline (RQ4)	1
Partial facial obstruction (surgical mask, eye patch, or wig)	1 or 0, reported separately*
Manipulated facial expression (smile, frown, or open mouth)	1 or 0, reported separately*
Full coverage face mask	0

The analysis will be analogous to that for **RQ2**, but with respect to liveness detection performance instead of age estimation performance. As with the analysis for **RQ2**, we will report

⁸ <https://developers.yoti.com/identity-verification/liveness>

⁹ <https://www.facetec.com/glossary>

confusion matrices, false positive rates, and false negative rates by study condition. We will also separately report analysis in which we combine null outputs with liveness outputs, as in the analysis for **RQ4**.

RQ6: How do demographic and personal attributes relate to liveness predictive performance?

We will carry out analysis similar to that for **RQ3**, using liveness detection rather than age estimation output. We will separately report results with combined null and liveness output as in the analysis for **RQ4**, and with differing ground truth for liveness, as in the analysis for **RQ5**.

RQ7: What is the joint age and liveness predictive performance of these services when people interact with them as directed?

In this analysis, we combine age estimation and liveness detection to simulate whether a person would be allowed to access an online service (d). In the typical configuration of online age assurance services, a user must both appear to meet an age threshold and pass liveness detection (when implemented). We define $d = e \wedge l$. When treating null outputs as equivalent to liveness failures, we define $d = e \wedge l \wedge \neg n$.

The analysis for this RQ will be analogous to the analysis for **RQ4**, with the addition of binary age estimation as in **RQ1**. Ground truth values are the same as in **RQ1**, since participants in the baseline condition are always live.

RQ8: How robust are joint age prediction and liveness detection to noncompliance with service directions or adversarial measures?

This analysis will be similar to that for **RQ5**, albeit with allowed status d instead of liveness l .

As ground truth for this analysis, we will continue with the binary age threshold ground truth from **RQ1**. This approach treats the policy judgment of who should be allowed access to an online service as the benchmark while accounting for the departures from vendor instructions (intentional or not) that can affect system output. So, for example, an adult who should be allowed access to an online service, and who passes an age check but fails a liveness check, would be treated as a false negative. Similarly, a minor who should not be allowed access to an online service, but who passes both age and liveness checks, would be treated as a false positive.

Specifically, we will use the following values for this analysis.

Table 7: Definitions of values for combined age estimation and liveness evaluation.

Value	Definition
True positive (TP)	The simulated user should be allowed access and is allowed access. The participant is of age, and the system reports that the user is of age and live (i.e., $a \geq t_e$ and $d = e \wedge l = 1$ or $d = e \wedge l \wedge \neg n = 1$, with the service configured to t_s).
True negative (TN)	The simulated user should not be allowed access and is not allowed access. The participant is not of age, and the system reports that the user is either not of age or not live (i.e., $a < t_e$ and $d = 0$, with the service configured to t_s).
False positive (FP)	The simulated user should be allowed access but is not allowed access. The participant is not of age, and the system reports that the user is of age and live (i.e., $a < t_e$ and $d = 1$, with the service configured to t_s).
False negative (FN)	The simulated user should not be allowed access and is allowed access. The participant is not of age, and the system reports that the user is either not of age or not live (i.e., $a < t_e$ and $d = 0$, with the service configured to t_s).

RQ9: How do demographic and personal attributes relate to combined age estimation and liveness predictive performance?

We will use the same approach as in the analysis for **RQ6**, but with the ground truth values and analysis values from **RQ8**.